SECTION 1

Time—30 minutes

40 QUESTIONS

For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. EXPEL: (A) await (B) adapt (C) loiter (D) disturb (E) admit

2. LENIENCY: (A) disdain (B) hostility (C) severity (D) persistence (E) stinginess

3. COMBUSTIBLE: (A) unlikely to break (B) unable to burn (C) reluctant to play (D) expected to fail (E) forced to open

4. DISAPPROVAL: (A) concern (B) security (C) cooperation (D) endorsement (E) tastelessness

5. SQUANDER: (A) agree (B) cleanse (C) conserve (D) treat fairly (E) speak eloquently

6. MULTIFARIOUS: (A) absorbent (B) clumsy (C) limited in variety (D) causing dissension (E) unintentionally harmful

7. PHILISTINE: (A) cultured (B) courageous (C) arrogant (D) generous (E) witty

8. BALEFUL: (A) vain (B) solid (C) benign (D) common (E) incessant

9. DISINTERMENT: (A) penetration (B) burial (C) denunciation (D) perseverance (E) construction

10. GLUT: (A) dearth (B) change (C) disparity (D) refinement (E) correction

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ---, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

11. Although most people have one explanation or another for what happened, the event remains ----.

(A) old-fashioned (B) unpopular (C) frightening (D) accidental (E) mysterious

12. The basic purpose of advertising is the announcement of products, prices, new equipment, and special sales and is, therefore, entirely ---- since such announcements ---- needed useful information.

(A) useless. .duplicate
(B) benevolent. .confirm
(C) suspect. .insinuate
(D) defensible. .eliminate
(E) reasonable. .provide

13. Good health is ---- with a high degree of resistance to bacterial attack; any influence that lowers one's general health also ---- one's resistance to such an attack.

(A) associated. .activates
(B) balanced. .secures
(C) allied. .increases
(D) synonymous. .reduces
(E) confused. .diminishes

GO ON TO THE NEXT PAGE

The reading passages in this test are brief excerpts or adaptations of excerpts from published material. To make the text suitable for testing purposes, we may in some cases have altered the style, contents, or point of view of the original.

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Princeton, N.J. 08541

14. MacDougall theorizes that in human history order has tended to arise from chaos and, similarly, that social forces will do almost anything to ---- anarchy.

(A) prolong (B) avert (C) instill
(D) imitate (E) instigate

15. Few politicians are so ---- asinine as to adopt a stance that is morally ---- and then boast of it.

(A) furtively. .precarious
(B) scrupulously. .laudable
(C) egregiously. .untenable
(D) surreptitiously. .tolerable
(E) transparently. .indisputable

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

16. OLDTIMER : NEWCOMER :: (A) host : party
(B) imitator : copy (C) veteran : inductee
(D) fable : legend (E) ancestor : relative

17. MARGIN : PAGE :: (A) envelope : letter
(B) crust : bread (C) grass : field
(D) tide : ocean (E) country : nation

18. DISMISS : EMPLOYEE :: (A) evict : tenant
(B) harass : enemy (C) dislike : food
(D) load : baggage (E) betray : friend

19. YARDSTICK : LENGTH :: (A) meter : capacity
(B) plateau : elevation (C) meridian : longitude
(D) hourglass : time (E) humidifier : moisture

20. SMILE : HAPPINESS :: (A) pallor : health
(B) laughter : pun (C) grimace : condolence
(D) frown : puzzlement (E) shout : contentment

21. LOW : CATTLE :: (A) wool : sheep
(B) fodder : horses (C) cluck : chickens
(D) nest : robins (E) litter : puppies

22. ABEYANCE : ACTIVITY :: (A) petition : request
(B) intermission : concert (C) exploit : adventure
(D) rebuttal : denial (E) rehearsal : drama

23. SANCTUARY : PROTECTION ::
(A) quagmire : difficulty
(B) labyrinth : circularity
(C) treadmill : variety
(D) apotheosis : ennui
(E) conundrum : lucidity

24. PALEONTOLOGY : FOSSILS ::
(A) philology : words (B) philosophy : adages
(C) oceanography : ships (D) meteorology : stars
(E) ontology : plants

25. LIMPID : MUDDINESS :: (A) eerie : fear
(B) ambiguous : doubt (C) iridescent : color
(D) chaotic : confusion (E) harmonious : discord

GO ON TO THE NEXT PAGE

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

(The passages for this test have been adapted from published material. The ideas contained in them do not necessarily represent the opinions of the College Board or Educational Testing Service.)

Compared with modern cities, cities of the Italian Renaissance were not large. In traveling from one side of the city to another, citizens could not avoid the main thoroughfares. They became accustomed to the same faces. In one way or another, citizens were always on show.

Citizens measured one another accordingly. There was no separation between the private and public lives of the citizens. The world of the individual was permeated with the values of the society. Parish and guild, family and neighborhood were the vessels from which the citizen drew identity and security.

We know too little about the values of the very poor. The record, such as it is, indicates that the poor constantly drifted in a quest for lower rents and higher wages. But, for the members of the tenacious and prosperous middle and upper classes, attachment to neighborhood and family was profound.

Hence, we can begin to understand why exile from one's native city was an agonizing experience. The effects of such estrangement were most acutely felt by members of the ruling class. They were the best known citizens in the city. They were born and bred to politics until the triumph of one-man rule transformed such grooming into a preparation for service to a prince.

Not surprisingly, the Aristotelian stress on people as political animals found an approving audience in the intelligentsia. Humanity seemed to have no reality outside a civil context.

26. Which of the following best summarizes the author's main point?

(A) Individual and social identity were one during the Italian Renaissance.
(B) Social class distinctions were at their height during the Italian Renaissance.
(C) Politicians of the Italian Renaissance were bred for their roles in the public sphere.
(D) The notion of man as a political animal was an Aristotelian ideal.
(E) Life in the cities of the Italian Renaissance was crowded and hectic.

27. According to the passage, exile from an Italian Renaissance city was an especially painful experience for members of the ruling class because they

(A) were poorly informed about customs in territories beyond their own city
(B) derived their identity from interaction with others in their community
(C) would have difficulty adjusting to an alien intellectual atmosphere
(D) had been taught to regard other princes as enemies
(E) would find it difficult to survive financially in a new place

28. It can be inferred that the history of the poor in the cities of the Italian Renaissance

(A) has been carefully suppressed
(B) is a record of discontent and rebellion
(C) deals mostly with their economic impact
(D) is chiefly an account of their migrations
(E) has been inadequately recorded

29. The passage is probably intended to

(A) argue an unpopular point of view
(B) present a nostalgic view of the period
(C) provide information about the cities
(D) suggest the weaknesses of the small city
(E) refute legends about the golden age of the Italian Renaissance

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1

One great group of seed-producing plants, the angiosperms (commonly known as flowering plants), is both the largest group in number of genera, species, and individuals, and the most recent to develop on earth. The angiosperms differ from gymnospermous seed plants in that their ovules and seeds are enclosed within the pistil, or megasporophyll, which later becomes a seed-bearing fruit. The structure of the pistil in certain primitive angiosperms suggests that the enclosure of the seeds may have come about by the evolutionary folding of a leaflike, ovule-bearing megasporophyll.

The angiosperms exceed all other vascular plants in range of diversity of plant body and habitat, and in their utility to mankind. Both woody and herbaceous angiosperms exist, and among the latter especially there is considerable diversity of vegetative structure, exemplified by bulbous hyacinths, onions, lilies, rhizomatous *Iris*, and many grasses. Diversity of habitat is demonstrated by such aquatics as water lilies, *Elodea*, and duckweed, or *Lemna*; such xerophytic genera as cacti; and such epiphytes as "Spanish moss," orchids, and bromeliads. In mature cacti, the leaves are either much reduced or absent.

Woody angiosperms are used extensively as lumber and fuel and as the source of commercial cork, while herbaceous types are important sources of food, beverages, textiles, drugs, and vegetable oils. Both the vegetative and reproductive portions of angiosperms are used as foods. Sweet potatoes, carrots, turnips, beets, and parsnips are examples of fleshy roots that are important foods; white or "Irish"potatoes and asparagus are stems. Various greens, such as spinach, turnip greens, chard, and lettuce are leaves. One eats the immense terminal buds of cabbages and head lettuce, whereas it is the fleshy petioles of rhubarb and celery that are eaten. Examples of the reproductive organs of angiosperms—flowers, fruits, and seeds—used as food are even more abundant. In both cauliflower and broccoli, we eat groups of flowers, called inflorescences. National cultures and economies are based on the use for food of such fruits as the grains of corn, rice, wheat, and rye. Indeed, fruits used as food are too numerous to list completely, but among them we may cite citrus fruits, squashes and melons, tomatoes, grapes, bananas, apples, pears, and various berries. The preceding list contains examples, such as tomatoes and squash, of fruits that laymen frequently classify as "vegetables."

30. According to the passage, which of the following can be inferred about squash?

(A) The edible parts of squash are called inflorescences.
(B) The edible parts of squash are derived from the reproductive organs of the plant.
(C) Like sweet potatoes, the edible parts of squash are roots.
(D) Squash is considered a woody angiosperm.
(E) Squash is often inaccurately classified as a fruit.

31. It can be inferred that which of the following was discussed in the paragraphs immediately preceding the passage?

(A) Plants that do not produce seeds
(B) The differences between woody and herbaceous angiosperms
(C) Blights that attack angiosperms
(D) Types of flowers produced by angiosperms
(E) Gymnospermous seed plants

32. The author mentions *Elodea*, cacti, and orchids primarily in order to emphasize the

(A) differences between woody and herbaceous angiosperms
(B) uses to which mankind has put angiosperms
(C) diversity of structure among angiosperms
(D) diversity of habitat among angiosperms
(E) similarity between xerophytic genera and epiphytes

33. The author's primary purpose in the passage is to

(A) give practical advice about growing angiosperms
(B) detail new discoveries in horticulture
(C) discuss the diversity of a particular group of plants
(D) question existing theories in botanical evolution
(E) trace evolutionary developments of gymnosperms

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People, once they have provided for their immediate needs, seem always to turn toward some form of creative expression. Yet strong feelings of guilt can often be found lurking somewhere in the background, based on many cultures' glorification of "practical" work. We need to know, I think, that our need to create is as valid as our need to maintain ourselves. If poetry (or any art form) has a function in society, it may be to serve as visible, audible evidence that creativity is a universal and timeless characteristic.

34. We can infer that the author would consider sculpture evidence of

(A) humanity's practical need for useful objects
(B) a means of escape from universal guilt feelings
(C) humanity's need to create
(D) realistic documentation of the differences in various cultures
(E) an entertaining diversion for those who must work at difficult tasks

35. According to the passage, people often feel guilty when they

(A) create works of art
(B) toil without enjoying it
(C) criticize those who are creative
(D) fail to help others meet their needs
(E) overemphasize the importance of practical tasks

Both conventional and continuous steelmaking begin with the melting of pig iron and scrap steel, together with fluxing, alloying, and purifying agents, in a furnace. It is in the shaping and solidifying of the molten metal that continuous casting offers its great economies in plant investment and in operation and energy costs. In the conventional procedure, the full ladle is moved to a casting pit where, supported by an overhead crane, it is tapped into a series of individual ingot molds standing upright on rail cars. When the surface of the ingots has solidified so that they can stand alone, the molds are stripped off. The ingots are then transferred to another furnace, the soaking pit, where they are heated to uniform temperature for the rolling to follow.

The hot ingots are first sent through the primary, or breakdown, mill: a series of massive rollers that squeeze the ingots down to the cross section appropriate to the type of product that is being made in that particular plant. The products of the primary mill are classified according to size and shape as slabs, blooms, and billets. Slabs have an oblong cross section; if they are destined to be rolled into plate or sheet, this cross section is usually larger than three by twelve inches. Blooms have a square cross section greater than five inches; billets are blooms that have been reduced to a smaller cross section.

Continuous-casting methods are not all the same, but they are enough alike to fit a generalized description. The ladle of molten steel is transported by a vehicle or crane to a platform above the casting machine, where it is teemed, or discharged, into a rectangular trough called a tundish. From the vessel the steel flows into a casting "strand," the heart of which is a bottomless vertical mold that receives and confines the molten metal, partially shaping it into the desired cross section as it passes through. Just below the upper surface of the metal a thin skin forms on the surfaces adjacent to the inner walls of the mold; this skin thickens as the metal is pulled through the mold.

The steel continues to solidify as the billet is withdrawn, and to hasten cooling it is usually passed through a secondary cooling chamber where it is sprayed with water from an array of nozzles. The solid billet, still at red heat, is pulled by rollers that are synchronized with the level of liquid metal in the mold in order to ensure that the billet is formed in a steady manner. Beyond the withdrawal mechanism the continuous billet is cut into convenient lengths for subsequent reduction. In short, continuous casting eliminates the pouring, molding, transporting, and reheating of ingots.

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36. The style of the passage can best be described as

(A) argumentative
(B) explanatory
(C) contemplative
(D) rhetorical
(E) scholarly

37. It can be inferred that steel is shaped into the final product at which stage of the casting process?

(A) When it has cooled in the molds in the casting pits
(B) When it has been formed into slabs, blooms, and billets
(C) When the billets have been cut and cooled completely
(D) Before it leaves the secondary cooling chamber
(E) After the stages described in the passage

38. The author provides information that answers which of the following questions?

I. In what ways does the continuous method differ from the conventional method of casting steel?
II. In what ways do the various methods of continuous casting differ from each other?
III. In what ways does the continuous-casting method provide economic benefit to the steel manufacturer?

(A) I only
(B) III only
(C) I and II only
(D) I and III only
(E) I, II, and III

39. According to the passage, in what way are the soaking pit and the secondary cooling chamber alike?

(A) In both, water is sprayed on the steel.
(B) In both, the steel is cooled and solidified.
(C) In both, steel is heated before passing through a series of rollers.
(D) Both receive steel in a partially solidified form.
(E) Both are stages in the conventional method of casting steel.

40. The passage is concerned primarily with

(A) presenting a complete picture of the ways in which steel is produced and molded into usable forms
(B) proving that there is little difference between the conventional and the continuous methods of steel production
(C) comparing the various methods of continuous casting of steel in order to present a generalized description
(D) tracing the development of the continuous-casting method of steel production as an outgrowth and modification of the conventional process
(E) demonstrating the advantages of continuous casting of steel by contrasting this process with the conventional method of steelmaking

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 2

Time—30 minutes

35 QUESTIONS

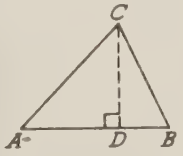
In this section solve each problem, using any available space on the page for scratchwork. Then indicate the one correct answer in the appropriate space on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definitions of symbols:

$=$	is equal to	$\leqq$	is less than or equal to
$\neq$	is unequal to	$\geqq$	is greater than or equal to
$<$	is less than	$\parallel$	is parallel to
$>$	is greater than	$\perp$	is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.
If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \dfrac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures which accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

---

1. If 2 out of every 5 students in a class of 35 intend to go to summer school, how many intend to go to summer school?

   (A) 5 (B) 7 (C) 10 (D) 14 (E) 21

2. If $2x = 5$ and $3y = 6$, then $6xy =$

   (A) 30 (B) 15 (C) 11 (D) $\frac{24}{5}$ (E) $\frac{12}{5}$

3. If $x + y = 6$ and $y + z = 9$, what is the value of z?

   (A) 2
   (B) 3
   (C) 6
   (D) 7
   (E) It cannot be determined from the information given.

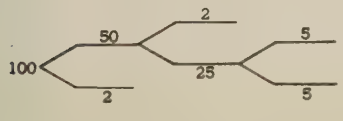


Figure I

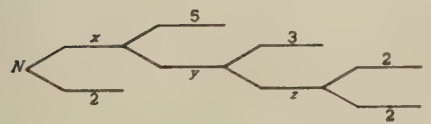
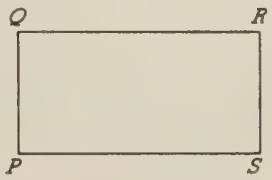


Figure II

4. Figure I is an example of a "factor diagram" of 100. What is the value of N if Figure II is a "factor diagram" of N?

   (A) 70 (B) 120 (C) 150
   (D) 240 (E) 300



5. In the rectangle above, $PQ = x$ and $QR = 2x$. What per cent of the perimeter of the rectangle is the sum $PQ + QR + RS$?

   (A) 50% (B) $66\frac{2}{3}\%$ (C) 75%
   (D) 80% (E) $83\frac{2}{3}\%$

---

6. If 45 cards can be copied in 30 minutes, how many hours will it take to copy 540 such cards at the same rate?

   (A) 3
   (B) 6
   (C) 12
   (D) 18
   (E) 24

7. Ms. Smith is S years old and is 3 years older than Ms. Lopez. In terms of S, how many years old was Ms. Lopez 2 years ago?

   (A) $S - 5$ (B) $S - 3$ (C) $S - 2$
   (D) $S - 1$ (E) $S + 1$

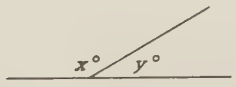
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Questions 8-27 each consist of two quantities, one in Column A and one in Column B. You are to compare the two quantities and on the answer sheet blacken space

A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

Notes: 1. In certain questions, information concerning one or both of the quantities to be compared is centered above the two columns.
2. A symbol that appears in both columns represents the same thing in Column A as it does in Column B.
3. Letters such as x, n, and k stand for real numbers.

EXAMPLES

	Column A	Column B	Answers
E1.	2×6	$2 + 6$	● Ⓑ Ⓒ Ⓓ
E2.	 $180 - x$	y	Ⓐ Ⓑ ● Ⓓ
E3.	$p - q$	$q - p$	Ⓐ Ⓑ Ⓒ ●

	Column A	Column B
8.	$\frac{6}{15}$	$\frac{11}{30}$

$y = x^2 - 1$
$z = w^2 - 1$

	Column A	Column B
9.	y when $x = 3$	z when $w = -3$



The total area of the rectangle is equal to the total area of the triangle. O is the center of the circle.

	Column A	Column B
10.	The area of the shaded region in the rectangle	The area of the shaded region in the triangle

	Column A	Column B
11.	3,560 rounded to the nearest hundred	3,546 rounded to the nearest hundred

$\frac{x}{2} = 15$

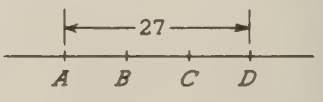
	Column A	Column B
12.	$\frac{x}{3}$	15

$x < y < 0$

	Column A	Column B
13.	x	xy

w, x, y, and z are integers.
$x \div 5 = y + 3$ and $w \div 5 = z + 2$

	Column A	Column B
14.	x	w



$AC = AB + 9$

$BC = \frac{BD}{2}$

	Column A	Column B
15.	AB	CD

The positive whole number divisors of 6 are 1, 2, 3, and 6.

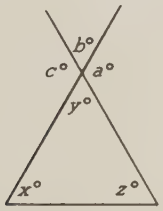
	Column A	Column B
16.	The number of positive whole number divisors of 20	5

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SUMMARY DIRECTIONS FOR COMPARISON QUESTIONS

Answer: A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

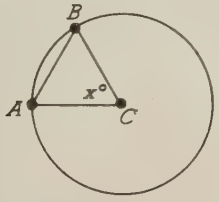
	Column A	Column B



17. $a+b+c$ | $x+y+z$

$2x+3=y$

18. $20x+20$ | $10y$



C is the center of the circle.

$x \neq 60$

19. $\frac{AB}{BC}$ | 1

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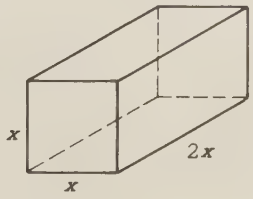
20. 1^k | 1^{2k}

Car	Speed (Kilometers per hour)	Time (Minutes)	Distance (Kilometers)
X	88 k.p.h.	x min.	1 kilometer
Y	90 k.p.h.	y min.	1 kilometer
Z	95 k.p.h.	z min.	1 kilometer

Each of the experimental cars X, Y, and Z maintains the constant speed indicated in the table above for the distance and time indicated.

21. x | z

	Column A	Column B



22. Total surface area of the rectangular solid shown | $10x^2$

---

23. The area of a right triangle with sides of lengths 3, 4, and 5 | The area of an isosceles triangle with two sides of length 5

$y=x+1$

24. The average (arithmetic mean) of 7, 9, and x | The average (arithmetic mean) of 2, 13, and y

---

$-1<x<1$

$x \neq 0$

25. $\frac{1}{x}$ | x

---

n is an odd integer greater than 125.

26. The remainder when $85n$ is divided by 2 | 1

$x<y<z$

27. xy | yz

GO ON TO THE NEXT PAGE

Solve each of the remaining problems in this section using any available space for scratchwork. Then indicate the one correct answer in the appropriate space on the answer sheet.

28. A right angle is bisected and each of the two resulting angles is trisected. Which of the following could NOT be the degree measure of an angle formed by any two of these lines?

(A) 15 (B) 30 (C) 50 (D) 60 (E) 75

29. If the product of 4 consecutive integers is zero, what is the least possible sum of these integers?

(A) −10 (B) −6 (C) −2 (D) −1 (E) 0

Questions 30-31 refer to the following table.

POPULATION PROJECTIONS BY REGION
1975-1985
(Thousands)

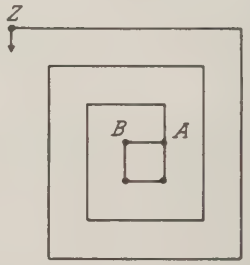
Region	1975 Population	Projected Population Change Between 1975 and 1985
P	2,671	202
Q	3,011	162
R	2,934	761
S	2,744	1,226
T	2,133	−134
U	2,210	631

30. What is the difference between the projected 1985 populations for regions T and U ?

(A) 497,000 (B) 574,000 (C) 765,000
(D) 842,000 (E) 923,000

31. Which of the six regions show(s) a projected population change for the period 1975-1985 of less than 10 per cent of the 1975 population?

(A) Q only
(B) P and S only
(C) Q and R only
(D) P, Q, and T only
(E) S, T, and U only



32. In the figure above, any two intersecting segments meet at right angles and any two adjacent parallel segments are a distance of 1 centimeter apart. What is the length of the solid-lined path from A to Z through B ?

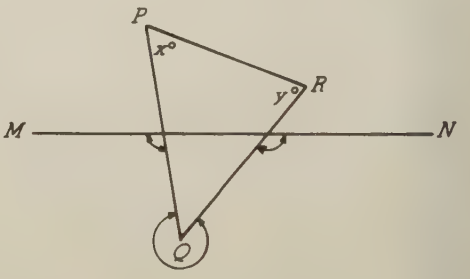
(A) 40 (B) 41 (C) 42 (D) 43 (E) 44

33. In a certain factory, $\frac{3}{4}$ of the workers are married and $\frac{3}{4}$ of these married workers have children. What fraction of the workers in the factory are married without children?

(A) $\frac{1}{16}$ (B) $\frac{3}{16}$ (C) $\frac{1}{4}$ (D) $\frac{1}{2}$ (E) $\frac{9}{16}$

34. If $x + y = k$ and $x - y = \frac{1}{k}$, then when $k \neq 0$, $x^2 - y^2 =$

(A) $\frac{1}{k}$ (B) k (C) k^2 (D) 2 (E) 1



35. In the figure above, what is the sum of the degree measures of the angles marked with arrows?

(A) $540 - 2x - 2y$
(B) $360 - 2x - 2y$
(C) 360
(D) 450
(E) 540

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 4

Time—30 minutes

45 QUESTIONS

For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. RUMPLED: (A) smooth (B) plentiful (C) recurrent (D) diverse (E) commonplace

2. CALAMITY: (A) pleasure (B) virtue (C) good fortune (D) suspicion (E) firm belief

3. OUTSPOKEN: (A) perceptive (B) reserved (C) ineligible (D) unnecessary (E) triumphant

4. REPRESS: (A) distract (B) annoy (C) borrow (D) release (E) miscalculate

5. CELESTIAL: (A) sectional (B) rational (C) earthly (D) realistic (E) feeble

6. TAPER: (A) make wider (B) make difficult (C) contradict (D) strengthen (E) remove

7. HUMID: (A) autumnal (B) polar (C) overcast (D) insipid (E) arid

8. DISBURSE: (A) oversee (B) proceed (C) collect (D) praise highly (E) gain weight

9. PRESTIGE: (A) lack of skill (B) lack of money (C) lack of influence (D) lack of ambition (E) lack of pride

10. ASCETIC: (A) parasitic (B) erratic (C) sensual (D) deleterious (E) merciless

11. RECANT: (A) avow (B) curse (C) restrain (D) deflate (E) compensate

12. TENUOUS: (A) endless or boring (B) extreme or excessive (C) firm or substantial (D) noisy (E) controlled

13. BILK: (A) banish fear (B) show respect for (C) portray faithfully (D) enforce strictly (E) deal honestly with

14. LACHRYMOSE: (A) dry-eyed (B) warmhearted (C) high-minded (D) sharp-sighted (E) wide-awake

15. COMMENSURATE: (A) unequal (B) stringent (C) improvident (D) inadvertent (E) immutable

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ---, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

16. Even though mistletoe berries are ----, they do contain medicinal properties and have been refined into a useful drug.

(A) scarce (B) hardy (C) colorful (D) poisonous (E) beneficial

17. In a sense, technology always ---- traditional social forms and creates a crisis for contemporary culture.

(A) underlies (B) disrupts (C) presupposes (D) delineates (E) facilitates

18. Maria's quick mind and ingenious imagination were evident in her ---- performances.

(A) reticent (B) marginal (C) banal (D) impromptu (E) stultifying

19. He was the scapegoat, by whose ---- punishment all the other transgressors in the group were, it is supposed, sufficiently ----.

(A) vicarious. .amused
(B) undisclosed. .avenged
(C) undeserved. .implicated
(D) sacrificial. .disillusioned
(E) public. .chastised

20. People of intemperate minds can never be free, for their passions ---- their fetters.

(A) forge (B) sever (C) shackle (D) tarnish (E) dissolve

GO ON TO THE NEXT PAGE

4

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

(The passages for this test have been adapted from published material. The ideas contained in them do not necessarily represent the opinions of the College Board or Educational Testing Service.)

Sometimes as I sit communing in my study I feel that death is not far off. Death neither alarms nor frightens one who has had a long career of fruitful toil. The knowledge that my work has been helpful to many fills me with joy and great satisfaction. Sometimes I ask myself if I have any legacy to leave. Truly, my worldly possessions are few. Yet, my experiences have been rich. From them, I have distilled principles and policies in which I believe firmly, for they represent the meaning of my life's work. Perhaps in them there is something of value. So, as my life draws to a close, I will pass them on to blacks everywhere.

I leave you the challenge of developing confidence in one another. Black banks, insurance companies, and other businesses are examples of successful economic enterprises. These institutions were made possible by vision and mutual aid. Economic separatism cannot be tolerated in this enlightened age, and it is not practicable.

I leave you a respect for the uses of power. We live in a world which respects power above all things. Power, intelligently directed, can lead to more freedom. We must select leaders who are wise, courageous, and of great moral stature and ability. We have great leaders among us today. We have had great men and women in the past: Frederick Douglass, Harriet Tubman, and Mary Terrell. These are people who worked not for themselves, but for others.

I leave you faith. The measure of our progress as a race is in precise relation to the depth of the faith in our people held by our leaders. Frederick Douglass was spurred by a deep conviction that his people would heed his counsel and follow him to freedom. Our forefathers struggled for liberty in conditions far more onerous than those we now face, but they never lost the faith. We must never forget their sufferings and sacrifices, for they were the foundations of the progress of our people.

I leave you racial dignity. We must recognize that we are the custodians as well as the heirs of a great civilization. We have given something to the world as a race and for this we are proud and fully conscious of our place in the total picture of mankind's development.

If I have a legacy to leave my people, it is my philosophy of living and serving, my vision of a world of Peace, Progress, Brotherhood, and Love.

21. The best title for the passage would be

(A) Thoughts of My Approaching Death
(B) A Legacy of Guiding Principles
(C) The Contributions of Black Leaders
(D) The Uses of Collective Power
(E) A Vision of Worldwide Peace

22. Which of the following statements is most compatible with the author's principles as presented in the passage?

(A) Too many leaders have rallied to wrong causes and destructive movements.
(B) Justice will prevail only in a world of cooperation and brotherhood.
(C) Stubborn perseverance sometimes hampers negotiations for liberation.
(D) Each individual must work for an abundant, secure personal life.
(E) Great leaders of the past could not have succeeded without formal education.

23. The author is primarily concerned with

(A) emphasizing the need for idealistic leaders
(B) exploring the possibilities of human nature
(C) portraying the present position of blacks in the United States
(D) presenting her life convictions to provide inspiration
(E) stating her views on the role of blacks in American history

24. It can be inferred from the passage that Mary Terrell (lines 28-29) was which of the following?

I. An intelligent and able leader
II. A pioneer in the use of economic power
III. An example because of her personal economic success

(A) I only (B) I and II only (C) I and III only (D) II and III only (E) I, II, and III

25. With which of the following statements regarding black leaders would the author be most likely to agree?

(A) Attempting to emulate past leaders is difficult because the times have changed greatly.
(B) Only leaders who have avoided personal fame should be held up as models.
(C) Past leaders provide more inspiration than present leaders.
(D) Only leaders who have achieved a measure of economic success should be imitated.
(E) If today's leaders lose faith in their race, our progress will be difficult or impossible.

GO ON TO THE NEXT PAGE

A characteristic of modern science is that it prefers to tackle well-defined, finite problems that appear to be soluble with the methods and evidence available. This often means eschewing the more fundamental, the more "metaphysical" issues, in the belief that the cumulative result of solving many smaller and more manageable problems will ultimately throw more light on the larger issues than would a frontal attack. One of the paradoxes of modern science has been that the greater its success in a pragmatic sense, the more modest its aims have tended to become in an intellectual sense. The goals and claims of modern quantum theory are far more modest than those of Laplace, who believed that he could predict the entire course of the universe, in principle, given its initial conditions.

The aim of science has changed from the "explanation" of reality to the "description" of reality—description with the greatest logical and aesthetic economy. The claims to universality of nineteenth-century physics have been replaced by a greater awareness of what still remains to be discovered about the world, even "in principle." The day of global theories of the social structure or of individual psychology seems to have passed. Experience has taught us that real insight has often been achieved only after we were prepared to renounce our claim that our theories were universal. The whole trend of modern scholarship has been towards greater conservatism in deciding what can be legitimately inferred from given evidence; we are more hesitant to extrapolate beyond the immediate circumstances to which the evidence applies. Even in art criticism, we are quicker to recognize a greater diversity in the influences playing on an artist, a greater ambiguity in his or her motives or artistic intentions.

The emphasis on finite aims in scientific and scholarly inquiry has been paralleled by the extension of scientific and scholarly attitudes to practical affairs. The preoccupation of science with manageable problems has a close analogy in the decline of ideology and the growth of professional expertise in politics and business. One of the most striking developments of the postwar world has been the increasing irrelevance of political ideology to political decision-making. One sees the influence of the new mood in the increasing bureaucratization of government and industry and in the growth of "scientific" approaches to management and administration. The day of the intuitive entrepreneur or the charismatic statesman seems to be waning.

26. The passage primarily concerns the

(A) diversity of issues for which modern scientists are seeking explanations
(B) means by which modern scientists have discredited nineteenth-century theories
(C) trend of modern scientists and scholars to focus on limited and manageable problems
(D) application of scientific methods to problems in political science and business
(E) challenges faced by modern scientists who are attempting to explain reality

27. The author's use of the phrase "a frontal attack" (line 9) could best be interpreted to mean which of the following?

(A) Finding practical applications for theories
(B) Questioning the validity of the available evidence
(C) Criticizing widely accepted theories
(D) Devising global theories of explanation
(E) Applying scientific methods to other fields of scholarship

28. According to the passage, modern scientists are more willing than their predecessors to

(A) recognize the limits of their knowledge
(B) find explanations for fundamental issues
(C) dismiss unresolved complexities and misunderstandings
(D) apply scientific methods to problems in other disciplines
(E) predict the future on the basis of the pattern of past events

29. The author sees a similarity between the concerns of modern science and which of the following trends in politics?

(A) A preoccupation among politicians with scholarly issues
(B) The reluctance of politicians to focus on practical issues
(C) Governmental resistance to the influence of industry
(D) The decreasing importance of political ideology
(E) The unpopularity of bureaucratic decision-making

30. The author apparently believes that the aim of science has changed from that of

(A) solving problems to that of devising questions
(B) analyzing causes to that of describing effects
(C) studying specific issues to that of inferring general solutions
(D) accepting experience to that of disregarding experience
(E) observing reality to that of interfering with reality

GO ON TO THE NEXT PAGE

Select the word or set of words that best completes each of the following sentences.

31. Virginia Woolf wrote fiction in which sensitive, ---- reactions to experience were objectified and patterned in a manner both intellectually exciting and aesthetically ----.

(A) impartial. .complete
(B) precise. .tranquil
(C) observant. .conclusive
(D) indefinite. .theoretical
(E) personal. .satisfying

32. Having lived in the public eye for years, the ex-mayor cannot believe that from now on he will live in ----.

(A) adulation (B) controversy (C) anonymity (D) agitation (E) veneration

33. Unlike Professor Stark, who is openly ---- about the value of the cell research, Dr. Gorman favors ---- the already sizeable funds allotted to the project.

(A) dubious. .trimming
(B) ecstatic. .donating
(C) enthusiastic. .endorsing
(D) skeptical. .augmenting
(E) curious. .scrutinizing

34. People were tired of what human ---- could provide; the invisible, the ---- were the need of the time.

(A) intellect. .irrational
(B) belief. .mysterious
(C) strength. .constructive
(D) beauty. .unheard
(E) beings. .traditional

35. Where one advances, hundreds ----; and the balance is always in favor of nearly universal ----.

(A) retreat. .peace
(B) battle. .weakness
(C) retrogress. .deterioration
(D) subside. .conflict
(E) stagnate. .subjugation

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep (B) anger : madness (C) smile : amusement (D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

36. CAPTAIN : TEAM :: (A) manager : responsibility (B) shortstop : basketball (C) teacher : book (D) administrator : staff (E) catcher : winner

37. INVITATION : SUMMONS ::
(A) procrastination : urgency
(B) action : thought
(C) request : demand
(D) suggestion : reminder
(E) telegram : cable

38. UNMASK : IDENTITY :: (A) concoct : plot (B) decode : message (C) escape : prison (D) submerge : emotion (E) exaggerate : truth

39. TADPOLE : AMPHIBIAN :: (A) lizard : insect (B) minnow : dolphin (C) lamb : mammal (D) branch : tree (E) rooster : chicken

40. GLOAT : SATISFACTION :: (A) gape : wonder (B) solve : mystery (C) glimpse : dismay (D) inspect : damage (E) disappear : surprise

41. DOMINEER : TYRANT :: (A) overrule : delegate (B) admire : champion (C) sympathize : victim (D) purchase : salesman (E) protect : guardian

42. CRIME : FELONY :: (A) worth : respect (B) wealth : gem (C) impiety : blasphemy (D) accident : injury (E) inferiority : rejection

43. STRIDENT : EAR :: (A) supple : skin (B) acrid : tongue (C) honeyed : lip (D) hypnotic : eye (E) saccharine : nose

44. GENRE : LITERARY :: (A) planet : astronomical (B) fallacy : philosophical (C) painting : artistic (D) phylum : zoological (E) plant : botanical

45. CULPABLE : CENSURE :: (A) moral : penance (B) meritorious : reward (C) laughable : abuse (D) admirable : judgment (E) affable : praise

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 5

Time—30 minutes

25 QUESTIONS

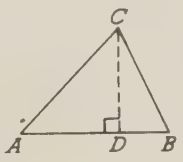
In this section solve each problem, using any available space on the page for scratchwork. Then indicate the one correct answer in the appropriate space on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area $= \pi r^2$; Circumference $= 2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definitions of symbols:

$=$	is equal to	$\leq$	is less than or equal to
$\neq$	is unequal to	$\geq$	is greater than or equal to
$<$	is less than	$\parallel$	is parallel to
$>$	is greater than	$\perp$	is perpendicular to

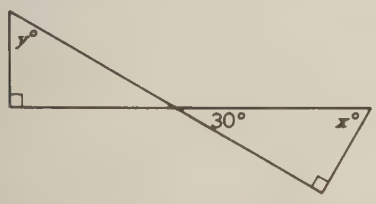


Triangle: The sum of the measures in degrees of the angles of a triangle is 180.
If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \frac{AB \times CD}{2}$

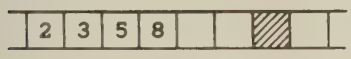
(2) $AC^2 = AD^2 + DC^2$

Note: Figures which accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.



1. In the figure above, $x - y =$

(A) 0 (B) 10 (C) 15 (D) 45 (E) 60



2. The boxes above show part of a sequence of numbers in which each number after 3 is the sum of the two numbers immediately to the left of it. If one number goes in each box, what number goes in the shaded box?

(A) 13 (B) 21 (C) 31 (D) 34 (E) 65

---

3. What is 99 per cent of 2 ?

(A) 0.0198
(B) 0.198
(C) 1.80
(D) 1.98
(E) 1.99

4. What is the result when $6 + 2x$ is subtracted from the sum of $2 - x$ and $3x^2 + 3x + 4$?

(A) $-3x^2$
(B) $-3x^2 - 2x + 4$
(C) $3x^2$
(D) $3x^2 + 2x + 6$
(E) $3x^2 + 6x + 8$

5. If $x^2 - 4 = 14 \cdot 18$, then x could be

(A) 12
(B) 14
(C) 16
(D) 18
(E) 20



6. A 7-hour clock is shown above. If at 12 noon today the pointer is at zero, where will the pointer be at 12 noon tomorrow?

(A) 0 (B) 2 (C) 3 (D) 5 (E) 6

GO ON TO THE NEXT PAGE

7. Of the following, which is greatest?

(A) $\frac{4}{7}$ (B) $\frac{3}{8}$ (C) $\frac{1}{3}$ (D) $\frac{2}{5}$ (E) $\frac{1}{2}$

8. If the average of the first 6 integers on a list is equal to the average of the first 5 integers on the list, then the sixth integer must be equal to

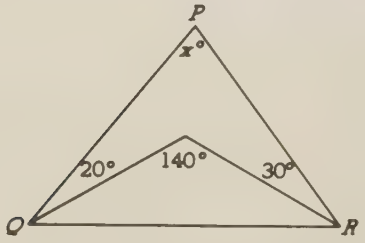
(A) the average of the first 5 integers
(B) an odd number
(C) a negative number
(D) the product of 5 and an integer
(E) the product of 6 and an integer

9. If $\sqrt{16} = x$, then $\sqrt{64} =$

(A) $4x$ (B) $2x$ (C) $x + 2$
(D) $2\sqrt{x}$ (E) $\sqrt{x}$

10. A worker is paid D dollars per hour per week for the first 40 hours worked and $2D$ dollars for each hour over 40. If a worker earns \$360 for one week's work, what is the value of D ?

(A) \$3 (B) \$5 (C) \$6 (D) \$9
(E) It cannot be determined from the information given.



Note: Figure not drawn to scale.

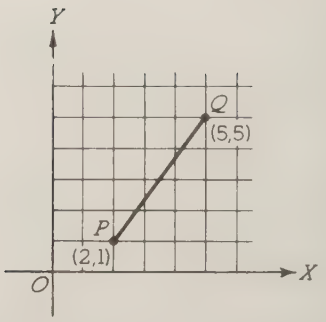
11. In $\triangle PQR$ above, $x =$

(A) 40 (B) 50 (C) 90 (D) 100 (E) 130

12. Which of the following can be expressed as the product of 3 different positive integers, each unequal to 1 ?

I. 24
II. 42
III. 45

(A) I only (B) I and II only (C) I and III only
(D) II and III only (E) I, II, and III

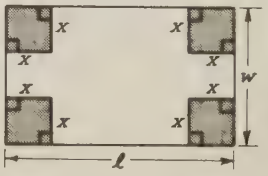


13. In the figure above, what is the length of PQ ?

(A) 13 (B) 7 (C) 6 (D) 5 (E) 3

14. Of the following values of N, $\left(-\frac{1}{3}\right)^N$ will be greatest for $N =$

(A) 2
(B) 3
(C) 4
(D) 5
(E) 6



15. In the figure above, the area of the unshaded region of the rectangle is

(A) $\ell w - 4x^2$
(B) $(\ell - 2x)(w - 2x)$
(C) $(\ell - x)(w - x)$
(D) $\ell x + wx$
(E) $(\ell - 4x)(w - 4x)$

GO ON TO THE NEXT PAGE

16. What is the diameter of a circle with circumference 1 ?

(A) π

(B) 1

(C) $\frac{1}{2}$

(D) $\frac{1}{\pi}$

(E) $-\pi + 1$

$X = 1 - \frac{9}{10}$

$Y = 1 - 0.99$

$Z = 1 \div 9$

17. If X, Y, and Z have the values indicated above, in which of the following are X, Y, and Z in order from least to greatest?

(A) X, Y, Z
(B) X, Z, Y
(C) Y, Z, X
(D) Y, X, Z
(E) Z, Y, X

18. A two-digit number has a tens' digit x and a units' digit y. What is the product of this number and the number 5, in terms of x and y ?

(A) $5x + y$
(B) $5x + 5y$
(C) $5x + 50y$
(D) $50x + 50y$
(E) $50x + 5y$

19. A 20-centimeter wire is cut into exactly three pieces. If the first piece is 3 centimeters shorter than the second piece and the third piece is 4 centimeters shorter than the second piece, what is the length in centimeters of the shortest piece?

(A) 5 (B) 6 (C) 7 (D) 8 (E) 9

20. If $0 < N < 1$, which of the following must be true?

I. $N > N^2$

II. $N > \frac{1}{N}$

III. $N > \frac{1}{\sqrt{N}}$

(A) I only (B) II only (C) III only
(D) I and II (E) I and III

21. A game board is made up of X rows of X squares. The squares of the board alternate in color from black to white and the corner squares are all white. If there are W white squares and B black squares, what is the value of $W - B$?

(A) -1
(B) 0
(C) 1
(D) 4
(E) It cannot be determined from the information given.

22. A child built a tower with a set of cubic blocks. If the first block had an edge of $2\frac{1}{2}$ inches and each block added had an edge which was $\frac{1}{4}$ inch less than the preceding block, what is the ratio of the volume of the fifth block to the volume of the first block?

(A) $\frac{1}{8}$

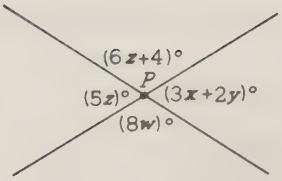
(B) $\frac{27}{125}$

(C) $\frac{9}{25}$

(D) $\frac{3}{5}$

(E) $\frac{125}{27}$

GO ON TO THE NEXT PAGE



Note: Figure not drawn to scale.

23. If the two straight lines in the figure above intersect at P, what is the value of w ?

(A) 10

(B) $12\frac{1}{2}$

(C) $17\frac{1}{2}$

(D) 20

(E) It cannot be determined from the information given.

24. If for all numbers n,
$\#n = n(n+1)(n+2)$, then $\frac{\#8}{\#4} =$

(A) $\#1$ (B) $\#2$ (C) $\#3$ (D) $\#4$ (E) $\#6$

25. In a certain bookcase 50 per cent of the books are math textbooks, 80 per cent of the books are over twenty years old, and 40 per cent of the books over twenty years old are history books. What is the maximum per cent of books in the bookcase that could be math textbooks over twenty years old?

(A) 28%
(B) 32%
(C) 40%
(D) 48%
(E) 50%

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

Correct Answers for Scholastic Aptitude Test
Form Code 0B023

VERBAL		MATHEMATICAL	
Section 1	Section 4	Section 2	Section 5
1. E	1. A	1. D	1. A
2. C	2. C	2. A	2. D
3. B	3. B	3. E	3. D
4. D	4. D	4. B	4. C
5. C	5. C	5. B	5. C
6. C	6. A	6. B	6. C
7. A	7. E	7. A	7. A
8. C	8. C	*8. A	8. A
9. B	9. C	*9. C	9. B
10. A	10. C	*10. C	10. E
11. E	11. A	*11. A	11. C
12. E	12. C	*12. B	12. B
13. D	13. E	*13. B	13. D
14. B	14. A	*14. D	14. A
15. C	15. A	*15. C	15. A
16. C	16. D	*16. A	16. D
17. B	17. B	*17. A	17. D
18. A	18. D	*18. B	18. E
19. D	19. E	*19. D	19. A
20. D	20. A	*20. C	20. A
21. C	21. B	*21. A	21. C
22. B	22. B	*22. C	22. B
23. A	23. D	*23. D	23. B
24. A	24. A	*24. C	24. A
25. E	25. E	*25. D	25. D
26. A	26. C	*26. C	
27. B	27. D	*27. D	
28. E	28. A	28. C	
29. C	29. D	29. B	
30. B	30. B	30. D	
31. E	31. E	31. D	
32. D	32. C	32. D	
33. C	33. D	33. B	
34. C	34. A	34. E	
35. A	35. C	35. E	
36. B	36. D		
37. E	37. C		
38. D	38. B		
39. D	39. C		
40. E	40. A		
	41. E		
	42. C		
	43. B		
	44. D		
	45. B		

*Indicates four-choice questions. (All of the other questions are five-choice.)

The Scoring Process

Machine-scoring is done in three steps:

- *Scanning.* Your answer sheet is "read" by a scanning machine and the oval you filled in for each question is recorded on a computer tape.
- *Scoring.* The computer compares the oval filled in for each question with the correct response. Each correct answer receives one point; omitted questions do not count toward your score. For each wrong answer, a fraction of a point is subtracted to correct for random guessing. For questions with five answer choices, one-fourth of a point is subtracted for each wrong response; for questions with four answer choices, one-third of a point is subtracted for each wrong response. For example, the SAT-verbal test has 85 questions with five answer choices each, and if a student has 44 right, 32 wrong, and 9 omitted, the resulting raw score is determined as follows:

  $$44 \text{ right} - \frac{32 \text{ wrong}}{4} = 44 - 8 = 36 \text{ raw score points}$$

  Obtaining raw scores frequently involves the rounding of fractional numbers to the nearest whole number. For example, a raw score of 36.25 is rounded to 36, the nearest whole number. A raw score of 36.50 is rounded upward to 37.
- *Converting to reported scaled score.* Raw test scores are then placed on the College Board scale of 200 to 800 through a process that adjusts scores to account for minor differences in difficulty between different editions of the test. This process, known as equating, is performed so that a student's reported score is not affected by the edition of the test taken nor by the abilities of the group with whom the student takes the test. As a result of placing ATP scores on the College Board scale, scores earned by students at different times can be compared. For example, an SAT-verbal score of 400 on a test taken at one administration indicates the same level of developed verbal ability as a 400 score obtained on a different edition of the test taken at another time.

How to Score the Test

You can verify the College Board SAT scores reported to you recently by using the information in this booklet along with the copy of your answer sheet. *Before you begin, check that the form code you marked in item 3 on your answer sheet is the same as the one printed on the front of this booklet.* Compare the responses shown on the copy of your answer sheet with the list of correct answers.

SAT-Verbal Sections 1 and 4

Step A: Count the number of correct answers for *section 1* and record the number in the space provided on the worksheet on the opposite page. Then do the same for the incorrect answers. (Do not count omitted answers.) To determine subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for *section 4* and record the numbers in the spaces provided on the worksheet. To determine subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: To obtain C, add subtotal A to subtotal B, keeping any decimals.

Step D: To obtain D, your raw verbal score, round C to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.)

Step E: To find your reported SAT-verbal score, look up the total raw verbal score you obtained in step D in the conversion table on the back cover.

SAT-Mathematical Sections 2 and 5

Step A: Count the number of correct answers and the number of incorrect answers for the *four-choice questions (questions 8 through 27) in section 2* and record the numbers in the spaces provided on the worksheet. To determine the subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{3} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for the *five-choice questions (questions 1 through 7 and 28 through 35) in section 2* and record the numbers in the spaces provided on the worksheet. To determine the subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: Count the number of correct answers and the number of incorrect answers for section 5 and record the numbers in the spaces provided on the worksheet. To determine the subtotal C, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal C}$$

Step D: To obtain D, add subtotal A, subtotal B, and subtotal C, keeping any decimals.

Step E: To obtain E, your raw mathematical score, round D to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.)

Step F: To find your reported SAT-mathematical score, look up the total raw mathematical score you obtained in E in the conversion table on the back cover.

SAT-Verbal Worksheet

A. Section 1: _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal A

B. Section 4: _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal B

C. Total unrounded raw score (Total A + B) _____ C

D. Total rounded raw score (Rounded to nearest whole number) _____ D

E. Reported SAT-verbal score (See the conversion table on the back cover.) [ ] SAT-Verbal

SAT-Mathematical Worksheet

A. Section 2: Questions 8 through 27 (4-choice) _____ (no. correct) − ⅓ (_____ no. incorrect) = _____ subtotal A

B. Section 2: Questions 1 through 7 and 28 through 35 (5-choice) _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal B

C. Section 5: _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal C

D. Total unrounded raw score (Total A + B + C) _____ D

E. Total rounded raw score (Rounded to nearest whole number) _____ E

F. Reported SAT-mathematical score (See the conversion table on the back cover.) [ ] SAT-Math

If you have any questions on these scoring instructions, you may call the phone number below.

If, after following the above scoring directions and checking your work carefully, your results disagree with the verbal or mathematical score reported on your ATP Student Report, you may request a hand scoring of your answer sheet by writing or calling:

College Board ATP
Box 592
Princeton, NJ 08541
Attention: Rescore Request
Telephone: (609) 883-8500

Please indicate whether it is your verbal or mathematical score, or both, that you want to be rescored. When you write, please include a copy of the scoring worksheet in this booklet on which you did your calculations.

Score Conversion Table
Scholastic Aptitude Test
Form Code 0B023

Raw Score	College Board Reported Score		Raw Score	College Board Reported Score	
	SAT-Verbal	SAT-Math		SAT-Verbal	SAT-Math
85	800		40	470	610
84	790		39	460	600
83	780		38	460	590
82	770		37	450	590
81	760		36	440	580
80	760		35	430	570
79	750		34	430	560
78	740		33	420	550
77	730		32	410	540
76	730		31	410	540
75	720		30	400	530
74	710		29	390	520
73	700		28	390	510
72	690		27	380	500
71	690		26	370	490
70	680		25	360	490
69	670		24	360	480
68	660		23	350	470
67	660		22	340	460
66	650		21	340	450
65	640		20	330	440
64	640		19	320	430
63	630		18	320	430
62	620		17	310	420
61	610		16	300	410
60	610	800	15	300	400
59	600	790	14	290	390
58	590	780	13	280	380
57	590	770	12	270	380
56	580	760	11	270	370
55	570	750	10	260	360
54	570	740	9	250	350
53	560	730	8	250	340
52	550	720	7	240	330
51	550	710	6	230	330
50	540	700	5	230	320
49	530	690	4	220	310
48	520	680	3	210	300
47	520	670	2	210	290
46	510	660	1	200	280
45	500	650	0	200	280
44	500	640	−1	200	270
43	490	640	−2	200	260
42	480	630	−3	200	250
41	480	620	−4	200	240
			−5	200	230
			−6	200	230
			−7	200	220
			−8	200	210
			−9 or below	200	200